Dag: a clearance gate for land registration

Stopping a bad deed before it is signed, instead of arguing about it for nine years.

Team	Silicon Knights	**Event**	LandTech 2026, Round 1
Problem	P4, Smart e-Mutation & Registry, with two additions under P6	**Date**	17 October 2026

1. THE PROBLEM WE PICKED

In Bangladesh a deed is registered at the sub-registry office under the Ministry of Law. The record of ownership is corrected at the AC (Land) office under the Ministry of Land. These are two ministries, two databases and two queues, and a deed registered in one of them does not necessarily reach the other [1]. The buyer walks out holding paper that the state does not yet recognise.

The government has already attacked half of this. e-Mutation cut the namjari process from about 60 days to 28 and from three or four office visits to one [2]. A registration-mutation interconnection was inaugurated in March 2023 and runs in a set of pilot upazilas [3]. On 1 January 2026 the Registration (Amendment) Ordinance amended the 1908 Act and opened the legal route for e-registration through government-approved software [4].

So the pipe between the two offices is being built. What nobody has built is the **gate**. Nothing today checks, at the counter, whether the deed about to be registered should be registered at all. That is the gap we are filling.

2. WHY IT MATTERS HERE

Roughly two million land cases sit with the judiciary, more than 70 percent of all litigation in the country [5]. The Senior Secretary of the Ministry of Land has put the figure near 80 percent [6]. One household in seven is in a land dispute, the average case runs about eight years, and some have run forty [5].

Most of these cases are not legal puzzles. They are bad entries that nobody checked. A single Prothom Alo investigation from Faridpur shows the whole chain in one story: after a landowner died, people claiming to be his heirs were brought in from across the border on forged certificates, the land was registered in their name, and almost immediately afterwards a third party used the same land as collateral for a bank loan [7].

> **That one case is the argument for both of our additions.** An alert to the real family would have exposed the registration within minutes. A charge check at the counter would have exposed the loan.

3. WHAT WE ARE BUILDING

Dag is a verification service that sits in front of registration. The sub-registrar enters the mouza, JL number, khatian, dag number, the seller, and the share being sold. Dag runs a fixed set of checks and returns a signed clearance: pass, warn, or block, with the reason and the evidence printed on it, and a QR code anyone can scan to confirm the clearance is real.

If the clearance passes and the deed is executed, Dag emits one event. That event opens the namjari case at the AC (Land) office with no application from the citizen, writes the transferred share into a ledger, and gives the citizen one tracking ID for the whole journey.

The name comes from the dag number, because the plot, not the person and not the document, is the only thing every office in the chain agrees on.

The seven checks

- **Seller in record.** Seller NID matches the current khatian holder.
- **Area ledger.** Shares already transferred out of the dag must not exceed the recorded area. This is what catches a 12-decimal plot being sold three times at 10 decimals.
- **Pending deed lock.** A deed in flight soft-locks the dag, so two offices cannot register the same plot in the same week.
- **Khajna and mutation.** Tax paid, and the seller holds a mutation khatian in his own name.
- **Government interest.** Khas, acquired, waqf, debottar, vested.
- **Court injunction.** Any pending suit recorded against the dag.
- **Survey chain.** CS to SA to RS to BS continuity for the dag, with a flag where the chain breaks.

Every block carries a reason code, the evidence behind it, and an appeal link. We added the appeal route deliberately. In Telangana, where registration is blocked automatically against a prohibited list, migration errors put thousands of legitimate private plots on that list and owners only discovered it at the counter [8]. A gate without an appeal path becomes a new kind of injustice.

4. THE TWO ADDITIONS (P6)

4.1 Owner Lock and Owner Alert

An owner registers his dag against his NID and mobile number. From then on, any deed, mutation application or mortgage touching that dag sends him an SMS within seconds. He can also set a lock. The lock does not stop a sale. It forces a second identity step and a 48-hour notice before the sub-registrar can proceed.

This is ordinary practice in United States county recorder offices and is now being made compulsory. Fifteen states require recorders to run owner alert systems, verify identity, or intervene when a suspicious document is filed; in Texas the recorder must notify the last known owner and refer the document to the district attorney [9]. From 1 January 2027 every California county must notify the owner whenever a deed or mortgage is recorded against his property [10]. Leavenworth County in Kansas and Platte County in Missouri go further and let an owner freeze his own deed [9].

Bangladesh has nothing of the kind. Every existing service is built around the buyer and the office. Nobody is watching on behalf of the person who already owns the land, which is exactly why a man working abroad, or a family that has just buried someone, finds out far too late. In Moulvibazar a Bangladesh Bank officer described losing control of family debutter property to forged signatures and a forged power of attorney, and said he could not even pay his land tax because false deeds had been filed at the land office [11].

4.2 Encumbrance oracle

Before registration, Dag asks the banking side one question: is there a live charge on this dag? It gets back yes or no, the date the charge was created, and the lender's name. Nothing else.

India solved this in 2011 with CERSAI, a central registry of security interests created so that a second lender can see the first lender's charge before granting a loan; the public can search it for ten rupees [12]. Bangladesh has the data but not the access. Bangladesh Bank's Credit Information Bureau runs a Collateral Information System, and since 2018 banks have been required to pull a collateral report before accepting a mortgage [13]. It is closed to citizens and to sub-registrars. The cost of that is visible: REHAB had received around 340 complaints of hidden developer mortgages by September last year, involving close to Tk 1,500 crore of buyers' money, and said openly that the country lacks a central digital registry of mortgaged property [14].

Our contribution is not the registry. It is the **yes/no oracle**. Bangladesh Bank cannot open CIB records to the public without amending the Bangladesh Bank Order [15], so a design that demands open data is a design that never ships. A signed yes/no discloses one bit, leaks no borrower information, and needs no change in the law.

5. SYSTEM ARCHITECTURE

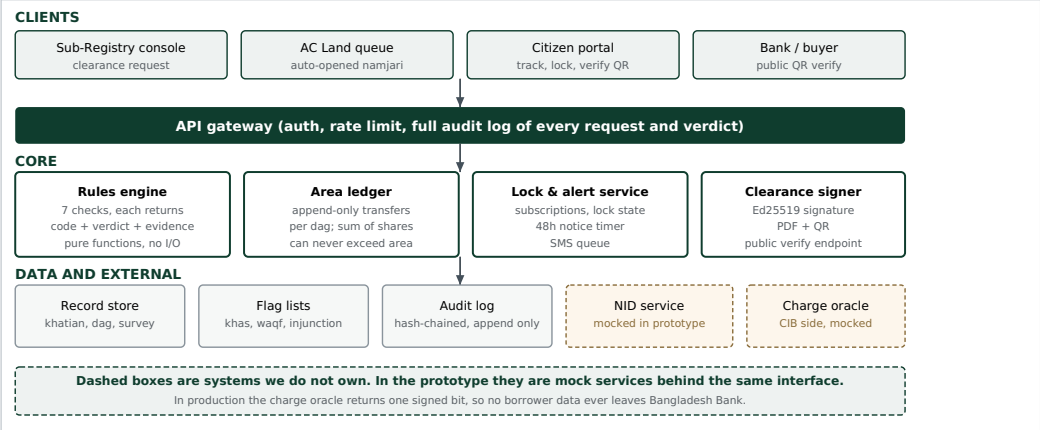


Figure 1. System architecture. Four layers, two mocked boundaries.

6. DECISION FLOW AT THE COUNTER

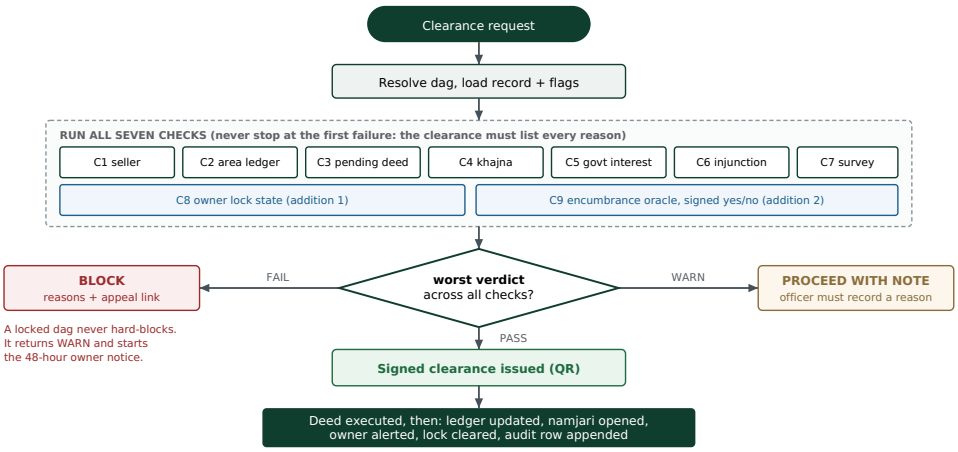


Figure 2. Decision flow. Blue boxes are the two additions. The gate reports every failed check, not the first one.

7. SEQUENCE OF ONE HONEST SALE, AND ONE ATTACK

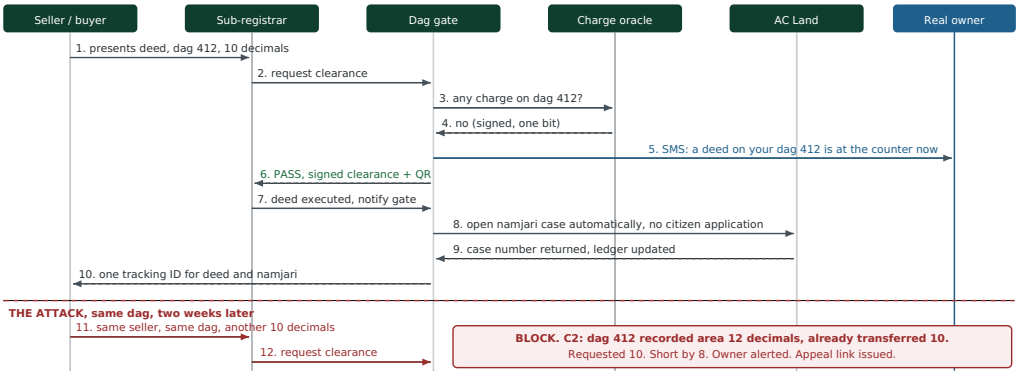


Figure 3. The honest sale clears in seconds. The second sale of the same land cannot clear, because arithmetic does not negotiate.

8. DATA PIPELINE FOR A MOUZA

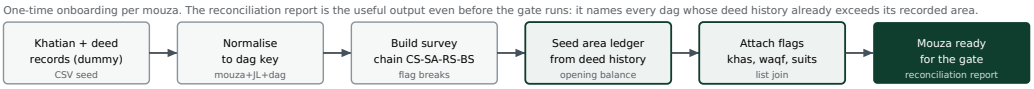


Figure 4. Onboarding pipeline. Run once per mouza.

9. TECH STACK

The prototype is a static web app: one link, no server to wake up. The gate's backend (rules engine, area ledger, lock-and-alert service, clearance signer, audit chain, and the mock NID, bank and SMS services) runs behind a single JSON API boundary, and every call is serialised both ways and shown live in the demo's API log. Clearances are signed with Ed25519 (TweetNaCl) over canonical JSON and carry a QR code; the audit log is a SHA-512 hash chain. The same backend files load in Node and are covered by 13 automated scenario tests, so moving them behind a real server with PostgreSQL changes one function, not the screens. We are not using a blockchain. The threat here is bad data entering a trusted registry, and a distributed ledger does not check whether a seller owns the land. A rules engine does.

10. NATIONWIDE APPLICABILITY

The gate is a service, not a replacement system. It sits between offices that already exist and speaks the identifiers those offices already use, so it can be switched on one upazila at a time, in read-only advisory mode first, and the reconciliation report alone gives the AC (Land) office a worklist on day one. Punjab in Pakistan ran this play and brought property registration down from 56 days to 11 across 55 million landowner records [16]. Karnataka made a verified digital record compulsory before registration, and revenue officials report that unauthorised registrations and the same-plot-sold-twice pattern stopped, while genuine registrations did not fall [17].

Beyond registration, the same clearance answers the questions the brief lists for other offices: banks checking collateral, acquisition offices identifying rightful owners before paying compensation, and courts needing a dated ownership history.

What we are not claiming

We did not invent the registration-to-mutation link; the government is already building it [3]. We are not proposing conclusive title. Mutation remains evidentiary, not determinative of ownership. And the charge oracle depends on a cooperation that does not exist yet. We have built our side of it and mocked the other side honestly, and the README says so.

11. TEAM AND MENTORS

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All data used in the prototype is invented. No real NID numbers, names, phone numbers or land records appear anywhere in the repository. AI tools used are disclosed in the README, and both students can explain every rule function and every line of the gate.

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